AMC 10 Simulation Test — Problems 1 - 13

25 Questions • 75 Minutes • Multiple Choice (A–E)

Instructions

- There are **25** problems. Choose the best answer **A–E** for each.
- Work carefully and keep your work neat. (No solutions are included in this document.)
- You may use scratch paper. A calculator is not needed.
- This test is restricted to problems that came up as problems 1 to 13 on the AMC 10.
- The problems are roughly ordered by difficulty.

1. One can holds 12 ounces of soda. What is the minimum number of cans to provide a gallon (128 ounces) of soda?

(A) 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 11

2. Marie does three equally time-consuming tasks in a row without taking breaks. She begins the first task at 1:00 PM and finishes the second task at 2:40 PM. When does she finish the third task?

(A) 3:10 PM **(B)** 3:30 PM **(C)** 4:00 PM **(D)** 4:10 PM **(E)** 4:30 PM

3. How many two-digit positive integers have at least one 7 as a digit?

(A) 10 **(B)** 18 **(C)** 19 **(D)** 20 **(E)** 30

4. Alicia earns \$20 per hour, of which 1.45% is deducted to pay local taxes. How many cents per hour of Alicia's wages are used to pay local taxes?

(A) 0.0029 **(B)** 0.029 **(C)** 0.29 **(D)** 2.9 **(E)** 29

5. How many three-digit positive integers have an odd number of even digits?

(A) 150 **(B)** 250 **(C)** 350 **(D)** 450 **(E)** 550

6. What is the value of x if $|x-1| = |x-2|$?

(A) $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** 1 **(D)** $\frac{3}{2}$ **(E)** 2

7. A digital watch displays hours and minutes with cAM and cPM. What is the largest possible sum of the digits in the display?

(A) 17 **(B)** 19 **(C)** 21 **(D)** 22 **(E)** 23

8. A rectangle with a diagonal of length x is twice as long as it is wide. What is the area of the rectangle?

(A) $\frac{1}{4}x^2$ **(B)** $\frac{2}{5}x^2$ **(C)** $\frac{1}{2}x^2$ **(D)** x^2 **(E)** $\frac{3}{2}x^2$

9. What is the sum of the digits of the square of $111,111,111$?

(A) 18 **(B)** 27 **(C)** 45 **(D)** 63 **(E)** 81

10. The sum of two positive numbers is 5 times their difference. What is the ratio of the larger number to the smaller?

(A) $\frac{5}{4}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{9}{5}$ **(D)** 2 **(E)** $\frac{5}{2}$

11. Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

(A) Sunday **(B)** Tuesday **(C)** Thursday **(D)** Friday **(E)** Saturday

12. A student must choose a program of four courses from a menu of courses consisting of English, Algebra, Geometry, History, Art, and Latin. This program must contain English and at least one mathematics course. In how many ways can this program be chosen?

(A) 6 **(B)** 8 **(C)** 9 **(D)** 12 **(E)** 16

13. Six points are equally spaced around a circle of radius 1. Three of these points are the vertices of a triangle that is neither equilateral nor isosceles. What is the area of this triangle?

(A) $\frac{\sqrt{3}}{3}$ **(B)** $\frac{\sqrt{3}}{2}$ **(C)** 1 **(D)** $\sqrt{2}$ **(E)** 2

14. In a certain year the price of gasoline rose by 20% during January, fell by 20% during February, rose by 25% during March, and fell by x% during April. The price of gasoline at the end of April was the same as it had been at the beginning of January. To the nearest integer, what is x?

(A) 12 **(B)** 17 **(C)** 20 **(D)** 25 **(E)** 35

15. A square has sides of length 10, and a circle centered at one of its vertices has radius 10. What is the area of the union of the regions enclosed by the square and the circle?

(A) $200 + 25\pi$ **(B)** $100 + 75\pi$ **(C)** $75 + 100\pi$ **(D)** $100 + 100\pi$ **(E)** $100 + 125\pi$

16. The state income tax where Kristin lives is levied at the rate of p% of the first \$28000 of annual income plus $(p + 2)$% of any amount above \$28000. Kristin noticed that the state income tax she paid amounted to $(p + 0.25)$% of her annual income. What was her annual income?

(A) \$28000 **(B)** \$32000 **(C)** \$35000 **(D)** $\$42,000$ **(E)** \$56000

17. A flower bouquet contains pink roses, red roses, pink carnations, and red carnations. One third of the pink flowers are roses, three fourths of the red flowers are carnations, and six tenths of the flowers are pink. What percent of the flowers are carnations?

(A) 15 **(B)** 30 **(C)** 40 **(D)** 60 **(E)** 70

18. One dimension of a cube is increased by 1, another is decreased by 1, and the third is left unchanged. The volume of the new rectangular solid is 5 less than that of the cube. What was the volume of the cube?

(A) 8 **(B)** 27 **(C)** 64 **(D)** 125 **(E)** 216

19. The first term of a sequence is 2005. Each succeeding term is the sum of the cubes of the digits of the previous terms. What is the 2005th term of the sequence?

(A) 29 **(B)** 55 **(C)** 85 **(D)** 133 **(E)** 250

20. How many 7 digit palindromes (numbers that read the same backward as forward) can be formed using the digits 2, 2, 3, 3, 5, 5, 5?

(A) 6 **(B)** 12 **(C)** 24 **(D)** 36 **(E)** 48

21. Which of the following expressions is never a prime number when p is a prime number?

(A) $p^2 + 16$ **(B)** $p^2 + 24$ **(C)** $p^2 + 26$ **(D)** $p^2 + 46$ **(E)** $p^2 + 96$

22. How many ordered pairs of real numbers (x, y) satisfy the following system of equations?

$$\begin{aligned} x + 3y &= 3 \\ \big||x| - |y|\big| &= 1 \end{aligned}$$

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** 8

23. How many positive integers n satisfy the following condition:

$$(130n)^{50} > n^{100} > 2^{200}?$$

(A) 0 **(B)** 7 **(C)** 12 **(D)** 65 **(E)** 125

24. Suppose that $P = 2^m$ and $Q = 3^n$. Which of the following is equal to 12^{mn} for every pair of integers (m, n)?

(A) P^2Q **(B)** P^nQ^m **(C)** P^nQ^{2m} **(D)** $P^{2m}Q^n$ **(E)** $P^{2n}Q^m$

25. A telephone number has the form $ABC - DEF - GHIJ$, where each letter represents a different digit. The digits in each part of the numbers are in decreasing order; that is, $A > B > C$, $D > E > F$, and $G > H > I > J$. Furthermore, D, E, and F are consecutive even digits; G, H, I, and J are consecutive odd digits; and $A + B + C = 9$. Find A.

(A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8